Beyond Computational Boundaries: Electromagnetic Field Reconstruction via FDTD and Boundary Integrals

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Short Background of the Problem

Electromagnetic waves propagate through unbounded space, but numerical simulations must be performed on finite computational domains. The boundaries made by human on computer may cause spurious reflections, and the cost of extending the computational domain to far-field observation points can be prohibitively expensive.

Hence we propose an alternative technique for computation of electromagnetic fields in open domains. Our strategy is to use finite-difference time-domain (FDTD) method with a proper boundary condition to solve Maxwell's equations in a bounded region that contains a given current source. Then extracting the electromagnetic field data on the enclosing surface and using a space-time Green's identity to reconstruct the fields at arbitrarily large distances, avoiding the need to discretize the entire region between source and observation locations.

Our steps are to develop the FDTD solver firstly and review boundary conditions and initial conditions. We will then discuss the formulation of Green's identity and its relationship to the time domain FDTD–BIE coupling.

Intended Approach and Computational Methods

- **Mathematical Model:** Consider Maxwell's equations in $\Omega = [-1, 1]^3$, with a prescribed current source $\mathbf{J}(\mathbf{x}, t)$ and appropriate initial and boundary conditions.
- **FDTD Simulation:** Implement the FDTD method[4] to compute the electromagnetic fields near the source and record field data on $\partial\Omega$.
- **Exterior Field Reconstruction:** Use the space-time Green's identity to reconstruct fields at distant observation points from the boundary data, without extending the FDTD grid.
- **Validation:** Compare FDTD results with a larger-domain reference simulation to evaluate boundary reflections and accuracy. Extend the reference domain to include the distant observation points, allowing direct comparison with the fields reconstructed using the space-time Green's identity. Examine numerical convergence and computational cost.
- **Further Investigation:** Explore coupling FDTD with time-domain boundary integral equations to provide nonreflecting boundary conditions, potentially developing a hybrid solver for open-domain electromagnetic simulations. Investigate its numerical stability, accuracy, and computational efficiency.

Metric of Success/Failure

We will evaluate the proposed method based on accuracy, convergence, stability, and computational efficiency; verify the reconstructed exterior fields with reference solutions obtained from larger-domain FDTD simulations that include the distant observation points directly. We will examine numerical convergence with grid and time-step refinement and investigate artificial boundary reflection effects. We will also compare run time and memory to assess the potential of our method to reduce computational cost at the expense of accuracy.

Finally, we will explore the potential of coupling FDTD with time-domain boundary integral equations to provide nonreflecting boundary conditions. We will examine the mathematical formulation and numerical challenges of this approach, particularly its stability, accuracy, and computational efficiency.

References

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